THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A sound wave has a frequency of 2000 Hz and a speed of 340 m/s. The number of waves in a 3.4 m length is: 1

(a) 2 **(b)** 10 **(c)** 20 **(d)** 40

2. A car travels half the distance at 40 km/h and the other half at 60 km/h. The average speed is: 1

(a) 50 km/h **(b)** 48 km/h **(c)** 52 km/h **(d)** 45 km/h

3. A person weighs 600 N on the Earth (g = 10). On a planet where g is one-third of that on Earth, the mass of the person is: 1

(a) 60 kg **(b)** 180 kg **(c)** 20 kg **(d)** 600 kg

4. A body of mass 4 kg is lifted through 3 m and then carried horizontally 5 m. The total work done against gravity is (g = 10): 1

(a) 320 J **(b)** 120 J **(c)** 200 J **(d)** 0

5. A wave of wavelength 2 m and speed 340 m/s has a time period of: 1

(a) 0.00588 s **(b)** 170 s **(c)** 680 s **(d)** 0.0059 Hz

6. A body in circular motion of radius 2 m makes 1 revolution in 4 s. Speed is (pi = 3.14): 1

(a) 3.14 m/s **(b)** 12.56 m/s **(c)** 1.57 m/s **(d)** 6.28 m/s

7. A block on a rough floor can resist a push of up to 12 N by static friction. If it is pushed by a force of 8 N and does not move, the friction force is: 1

(a) 8 N **(b)** 12 N **(c)** 4 N **(d)** 0

8. A 60 W lamp is used for 5 hours daily for 30 days. At 5 rupees per unit the cost of the energy is: 1

(a) 225 rupees **(b)** 45 rupees **(c)** 450 rupees **(d)** 9 rupees

9. If the amplitude of a sound wave is doubled, the loudness: 1

(a) increases **(b)** becomes zero **(c)** stays same **(d)** decreases

10. A car brakes from 20 m/s to rest with retardation 5 m/s squared. The distance covered is: 1

(a) 100 m **(b)** 10 m **(c)** 40 m **(d)** 4 m

11. A 0.5 kg ball moving at 20 m/s is stopped in 0.1 s by a player. The force on the hand is: 1

(a) 50 N **(b)** 10 N **(c)** 1000 N **(d)** 100 N

12. In a lever the effort arm is 40 cm and the load arm is 10 cm. The effort needed to lift a 200 N load is: 1

(a) 20 N **(b)** 800 N **(c)** 50 N **(d)** 200 N

13. In a sound wave, the distance between a compression and the next rarefaction is: 1

(a) half a wavelength **(b)** zero
(c) two wavelengths **(d)** one wavelength

14. A graph of velocity against time is a straight line passing through the origin. The body has: 1

(a) uniform acceleration from rest **(b)** uniform velocity
(c) zero acceleration **(d)** retardation

15. A block is pulled with a force of 15 N and moves with constant velocity. The friction on it is: 1

(a) 30 N **(b)** 7.5 N **(c)** 15 N **(d)** 0

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the force needed to stop a 2 kg body moving at 10 m/s in 4 s. 2

17. A bullet of mass 10 g moving at 300 m/s penetrates 3 cm into a wall. Find the average resistive force. 2

18. Why is the voice of a person who has inhaled helium squeaky? 2

19. Why is the free fall of a feather and a stone in a vacuum identical? 2

20. Why does a passenger in a car feel pushed to the right when the car turns left? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Calculate the time interval between a lightning flash and the thunder if a lightning strikes 3.4 km away (v = 340 m/s). 3

22. An object has a uniform velocity of 8 m/s for 5 s, rests for 5 s and then moves with 4 m/s for 10 s. Find the total distance and the average speed. 3

23. A hose pipe shoots water at 5 m/s at the rate of 2 kg/s on a wall. Find the force on the wall if the water stops. 3

24. Explain why the energy of a falling body is converted to heat when it hits the ground. 3

OR

Explain the working of a megaphone and a hearing aid. 3

25. A ball is thrown upward with 25 m/s. Find the time to rise, the maximum height, and the velocity at 1 s (g = 10). 3

OR

Explain, with a numerical example, how the time of impact affects the force: a 0.2 kg ball at 30 m/s is stopped in 0.02 s and in 0.2 s. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A driver at 72 km/h sees an obstacle 50 m ahead and brakes with a retardation of 5 m/s squared after a reaction time of 0.5 s. Will he stop in time? 5

OR

Explain the origin of friction, the factors on which it depends, and calculate the friction force on a 20 kg body with a coefficient of friction 0.3 (g = 10) and the acceleration if pulled with 80 N. 5

27. Explain the efficiency of a machine. A machine is supplied with 800 J and does 600 J of useful work. Find the efficiency and the energy lost. 5

OR

Explain, with graphs described in words, how two sounds with different amplitudes and frequencies differ. Sketch the case for A (amplitude 2, frequency 100 Hz) and B (amplitude 1, frequency 200 Hz). 5

END OF QUESTION PAPER **Total: 50 Marks**

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